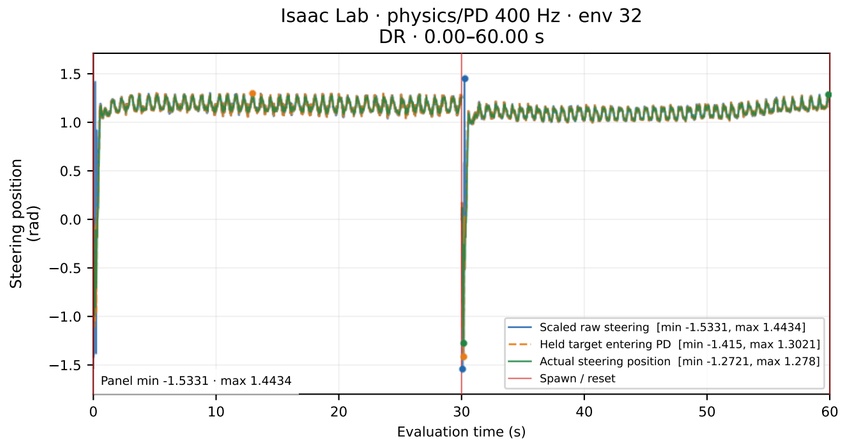
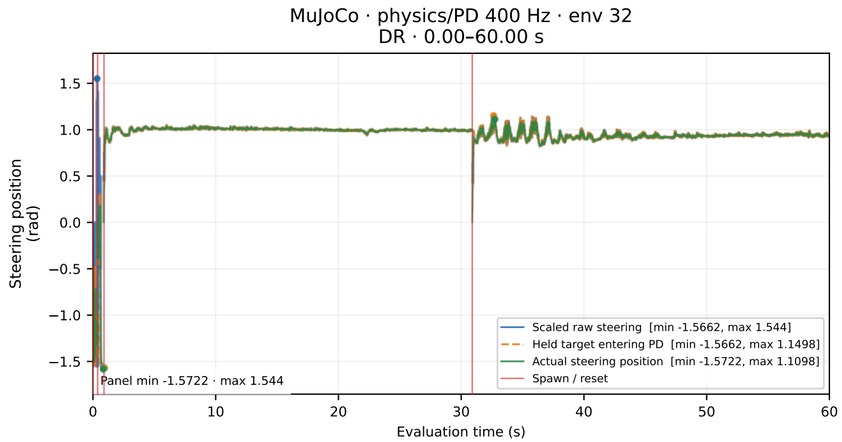
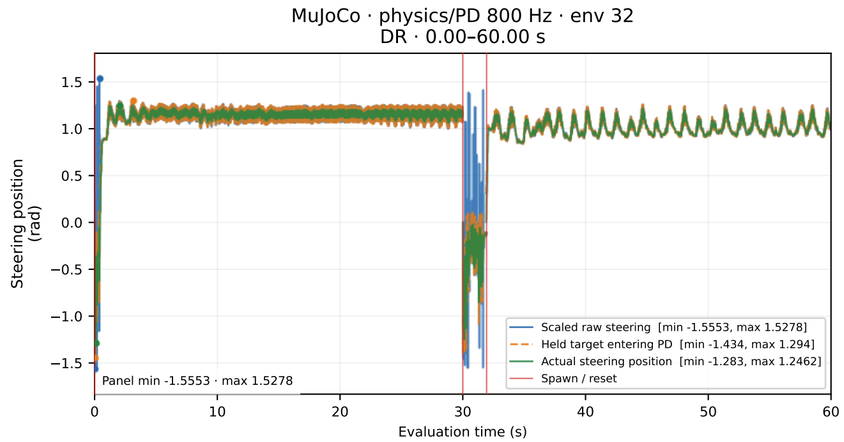
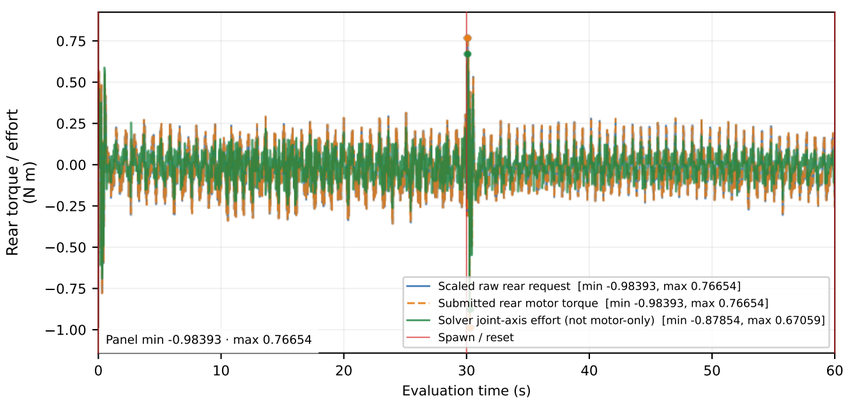
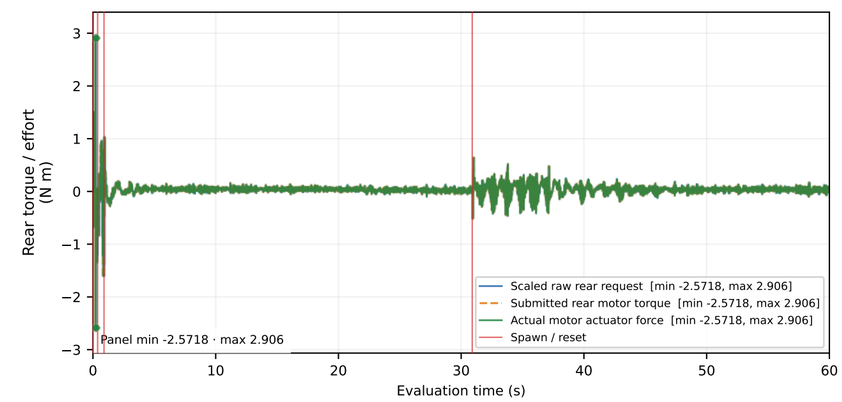
model_2500 · Demand versus actuation · env 32 · full 60 s

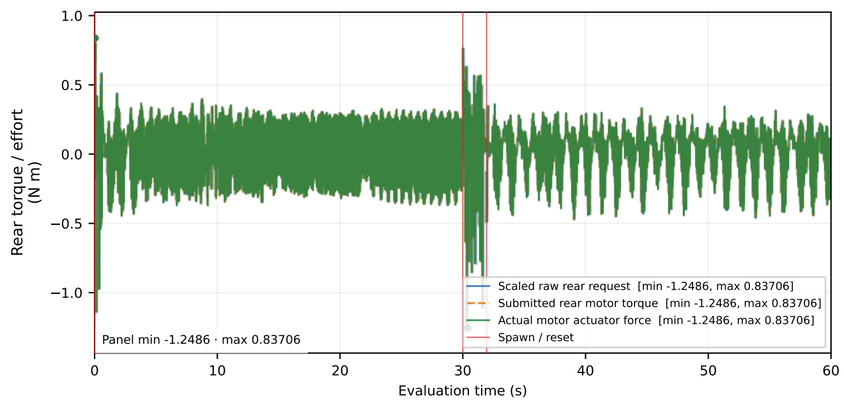












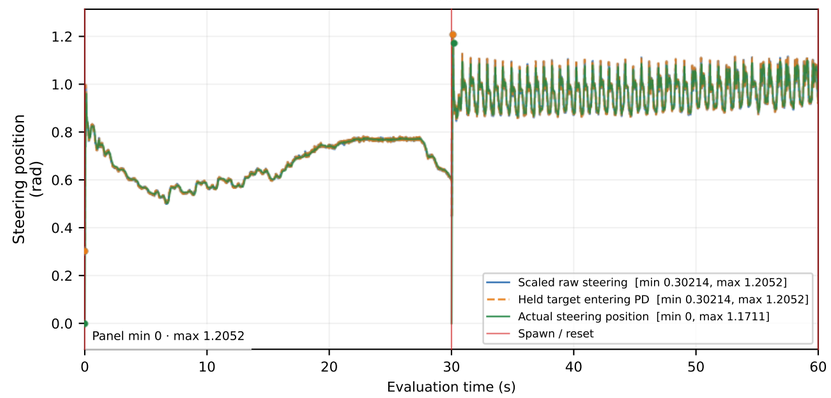
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

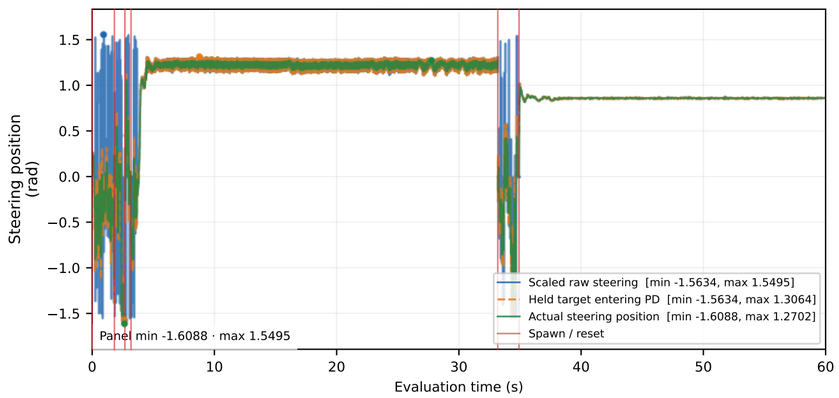
```
SAMPLED DR: env 32; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.15354, 0.994755, 1.13212, 0.895552]
  CoM XYZ mm [0.827014, -0.963841, -4.22029]
  mu_s [0.533455, 0.794338, 0.659445, 0.628821]
  mu_d [0.533455, 0.526209, 0.547619, 0.628821]
  Kp [steer,rear] [9.07548, 0]; Kd [0.198498, 0]
  spawn roll -0.0426314 rad; rate -0.19918 rad/s
  gyro bias XYZ rad/s [-0.000229271, 0.00035585, 0.000120208]
  gravity bias YZ [-0.00213508, -0.0010165]
E2 [30.00, 60.00) s:
  mass x [1.11867, 1.17444, 0.927634, 0.801768]
  CoM XYZ mm [2.46421, -1.17602, 4.30979]
  mu_s [0.741039, 0.764681, 0.559809, 0.592046]
  mu_d [0.614405, 0.764681, 0.559809, 0.534947]
  Kp [steer,rear] [9.73381, 0]; Kd [0.202134, 0]
  spawn roll -0.0633899 rad; rate 0.130295 rad/s
  gyro bias XYZ rad/s [-0.000184547, -0.000195907, 0.000120958]
  gravity bias YZ [0.000194793, -8.68671e-05]
Six significant digits shown; full precision: sampled_dr_values.json.
```

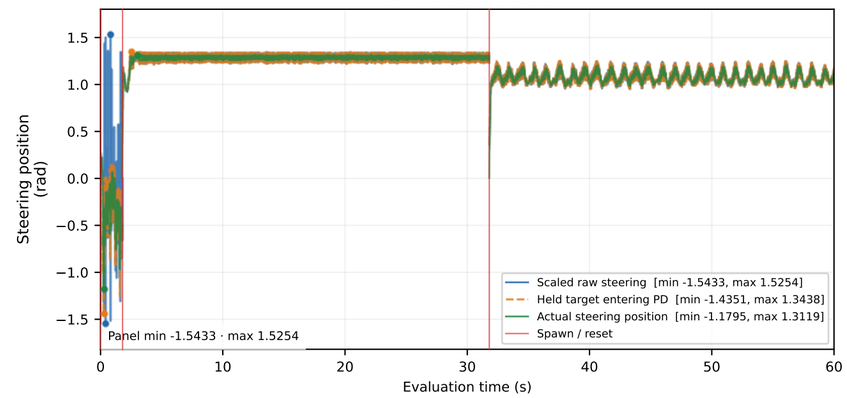
```
SAMPLED DR: env 32; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.38) s:
  mass x [1.15775, 1.00829, 1.07056, 1.06768]
  CoM XYZ mm [0.648381, 0.245347, -4.96027]
  mu_s [0.669571, 0.616244, 0.723429, 0.669571]
  mu_d [0.669571, 0.586498, 0.723429, 0.669571]
  Kp [steer,rear] [10.5561, 0]; Kd [0.218481, 0]
  spawn roll -0.0797969 rad; rate -0.148618 rad/s
  gyro bias XYZ rad/s [0.000199814, -6.26873e-05, 1.47468e-05]
  gravity bias YZ [0.000709582, 0.000497162]
E2 [0.38, 0.90) s:
  mass x [1.16707, 0.822879, 1.17134, 0.853268]
  CoM XYZ mm [-4.69558, -2.80143, 0.637254]
  mu_s [0.541939, 0.637675, 0.568073, 0.546287]
  mu_d [0.534359, 0.637675, 0.568073, 0.546287]
  Kp [steer,rear] [10.1914, 0]; Kd [0.19981, 0]
  spawn roll -0.04112596 rad; rate -0.0897145 rad/s
  gyro bias XYZ rad/s [0.00037647, 7.1709e-05, -3.5109e-05]
  gravity bias YZ [0.00389114, -0.00101151]
E3 [0.90, 30.90) s:
  mass x [1.17554, 1.16043, 0.862716, 0.958754]
  CoM XYZ mm [1.08892, 1.02467, 3.20079]
  mu_s [0.723429, 0.757579, 0.69316, 0.568073]
  mu_d [0.723429, 0.70921, 0.69316, 0.568073]
  Kp [steer,rear] [10.1718, 0]; Kd [0.19536, 0]
  spawn roll 0.0820112 rad; rate 0.0157743 rad/s
  gyro bias XYZ rad/s [-9.90621e-05, -0.00010471, 0.000284694]
  gravity bias YZ [0.00247652, 0.00237624]
E4 [30.90, 60.00) s:
  mass x [0.876745, 0.875549, 0.953014, 0.9418]
  CoM XYZ mm [-2.51835, 2.13953, -3.58757]
  mu_s [0.791209, 0.640867, 0.669571, 0.704749]
  mu_d [0.767936, 0.556841, 0.669571, 0.541926]
  Kp [steer,rear] [10.1907, 0]; Kd [0.194124, 0]
  spawn roll 0.0385587 rad; rate 0.0718184 rad/s
  gyro bias XYZ rad/s [-0.000119636, 0.000344695, 5.95735e-05]
  gravity bias YZ [0.000369942, 0.000484901]
Six significant digits shown; full precision: sampled_dr_values.json.
```

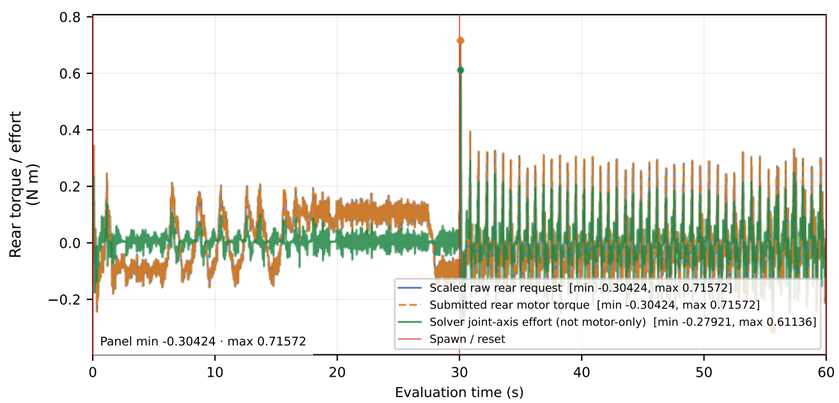
```
SAMPLED DR: env 32; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.15775, 1.00829, 1.07056, 1.06768]
  CoM XYZ mm [0.648381, 0.245347, -4.96027]
  mu_s [0.669571, 0.616244, 0.723429, 0.669571]
  mu_d [0.669571, 0.586498, 0.723429, 0.669571]
  Kp [steer,rear] [10.5561, 0]; Kd [0.218481, 0]
  spawn roll -0.0797969 rad; rate -0.148618 rad/s
  gyro bias XYZ rad/s [0.000199814, -6.26873e-05, 1.47468e-05]
  gravity bias YZ [0.000709582, 0.000497162]
E2 [30.00, 31.94) s:
  mass x [0.844711, 1.06078, 1.07912, 1.0883]
  CoM XYZ mm [-0.750851, 2.08231, 3.34438]
  mu_s [0.611239, 0.791209, 0.538976, 0.748289]
  mu_d [0.611239, 0.767936, 0.538976, 0.689499]
  Kp [steer,rear] [10.4035, 0]; Kd [0.215361, 0]
  spawn roll -0.0623382 rad; rate -0.11613 rad/s
  gyro bias XYZ rad/s [-6.18384e-05, -0.000112417, 0.000212459]
  gravity bias YZ [-0.000220083, -0.00215042]
E3 [31.94, 60.00) s:
  mass x [0.918723, 0.858997, 1.01245, 1.1257]
  CoM XYZ mm [-3.37096, 1.56511, -1.90493]
  mu_s [0.727426, 0.69316, 0.71008, 0.732187]
  mu_d [0.606358, 0.69316, 0.59371, 0.631664]
  Kp [steer,rear] [10.2083, 0]; Kd [0.199927, 0]
  spawn roll 0.00198396 rad; rate 0.0446447 rad/s
  gyro bias XYZ rad/s [-0.000102384, -0.000182165, 0.000137707]
  gravity bias YZ [0.00167768, -0.00296171]
Six significant digits shown; full precision: sampled_dr_values.json.
```

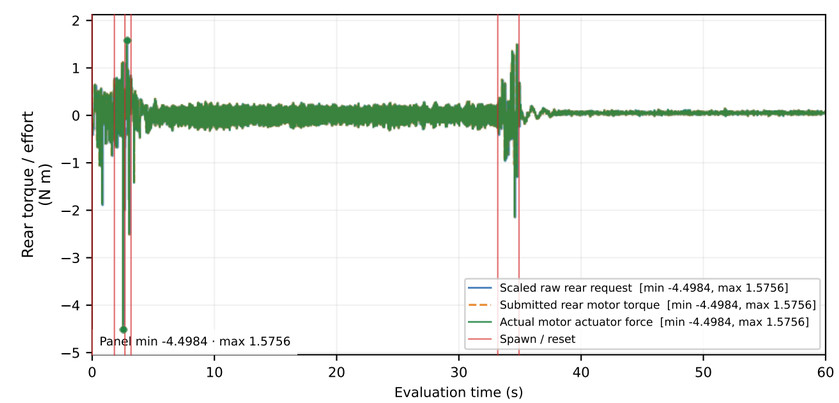
model_2500 · Demand versus actuation · env 33 · full 60 s

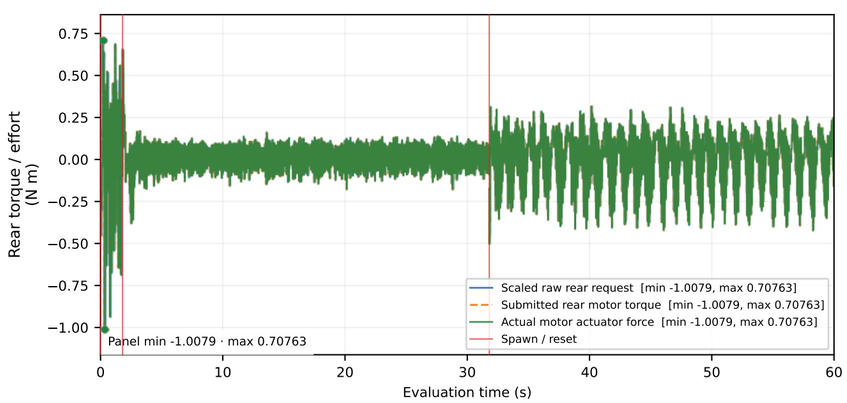












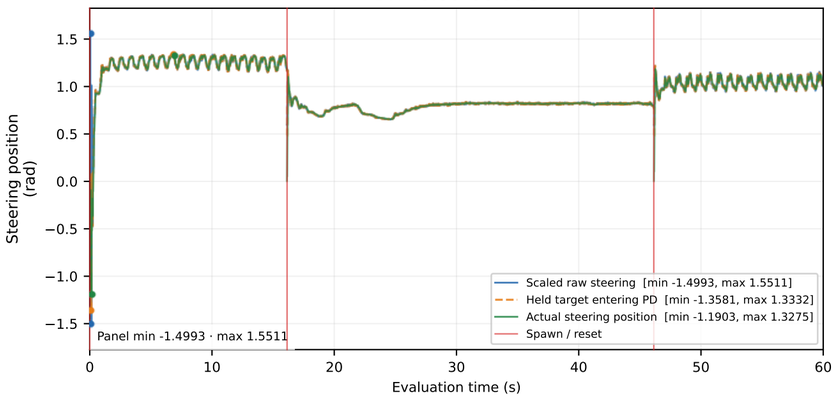
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

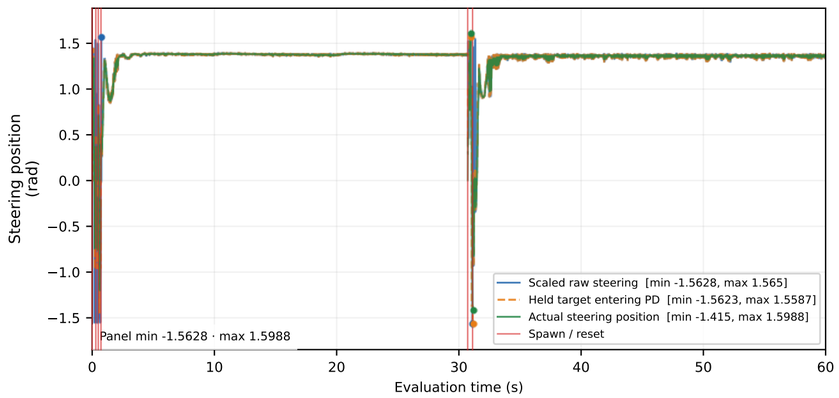
```
SAMPLED DR: env 33; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.918787, 0.909582, 0.860474, 0.816054]
  CoM XYZ mm [-1.41342, 4.90098, 1.82721]
  mu_s [0.547743, 0.547743, 0.674963, 0.674963]
  mu_d [0.547743, 0.547743, 0.601294, 0.601294]
  Kp [steer,rear] [9.07892, 0]; Kd [0.187439, 0]
  spawn roll 0.0479791 rad; rate -0.107307 rad/s
  gyro bias XYZ rad/s [5.93858e-05, 0.000281976, 0.000140135]
  gravity bias YZ [-0.000156539, 0.00012416]
E2 [30.00, 60.00) s:
  mass x [1.18798, 0.981603, 0.839946, 1.0543]
  CoM XYZ mm [-2.37614, 1.46068, 1.5182]
  mu_s [0.532645, 0.751971, 0.547743, 0.632409]
  mu_d [0.532645, 0.541224, 0.547743, 0.589076]
  Kp [steer,rear] [9.99212, 0]; Kd [0.192678, 0]
  spawn roll 0.0498407 rad; rate -0.0544084 rad/s
  gyro bias XYZ rad/s [0.000215835, 0.000132055, -0.000148476]
  gravity bias YZ [0.00101097, -0.00239106]
Six significant digits shown; full precision: sampled_dr_values.json.
```

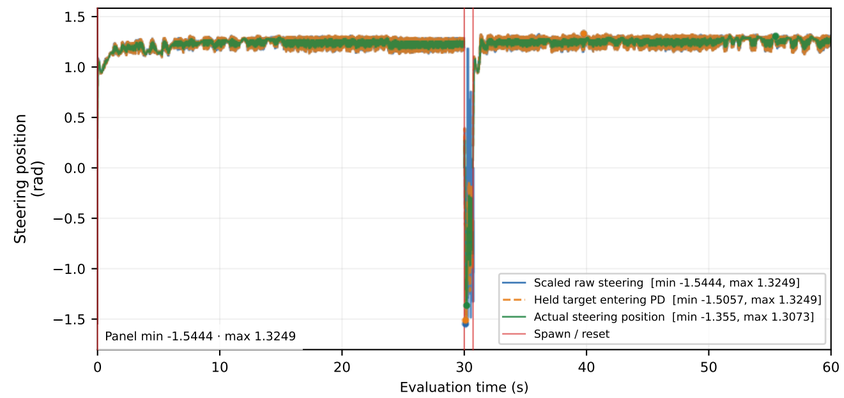
```
SAMPLED DR: env 33; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 1.82) s:
  mass x [0.885317, 0.861252, 1.0657, 1.07373]
  CoM XYZ mm [4.33164, 2.5149, 4.26862]
  mu_s [0.704749, 0.700521, 0.749678, 0.71155]
  mu_d [0.541926, 0.641329, 0.741429, 0.71155]
  Kp [steer,rear] [9.0252, 0]; Kd [0.212922, 0]
  spawn roll -0.0197365 rad; rate 0.0835653 rad/s
  gyro bias XYZ rad/s [-3.26169e-05, 0.000319795, -0.000258576]
  gravity bias YZ [-0.00102961, -0.00178678]
E2 [1.82, 2.68) s:
  mass x [0.977003, 0.802712, 0.882463, 0.834067]
  CoM XYZ mm [-2.07259, -2.23026, -0.33118]
  mu_s [0.611239, 0.71008, 0.704749, 0.538976]
  mu_d [0.611239, 0.59371, 0.541926, 0.538976]
  Kp [steer,rear] [10.9342, 0]; Kd [0.192036, 0]
  spawn roll -0.0112658 rad; rate 0.0655895 rad/s
  gyro bias XYZ rad/s [0.000292643, -0.000192386, 0.000102021]
  gravity bias YZ [-0.00345355, 0.000951618]
E3 [2.68, 3.18) s:
  mass x [0.99145, 1.09962, 0.923818, 0.832726]
  CoM XYZ mm [2.22311, 1.26657, -4.71255]
  mu_s [0.757579, 0.640016, 0.669571, 0.568073]
  mu_d [0.70921, 0.513141, 0.669571, 0.568073]
  Kp [steer,rear] [9.43703, 0]; Kd [0.212337, 0]
  spawn roll 0.0622928 rad; rate -0.0996375 rad/s
  gyro bias XYZ rad/s [0.000216561, 0.000179249, 0.000152826]
  gravity bias YZ [-0.000605763, -0.00251901]
E4 [3.18, 33.18) s:
  mass x [0.951043, 1.06067, 1.1946, 1.08842]
  CoM XYZ mm [-2.4262, -1.6443, 0.533732]
  mu_s [0.757579, 0.728342, 0.71008, 0.597748]
  mu_d [0.70921, 0.728342, 0.59371, 0.597748]
  Kp [steer,rear] [9.39824, 0]; Kd [0.181749, 0]
  spawn roll -0.0559239 rad; rate -0.19513 rad/s
  gyro bias XYZ rad/s [0.000296256, -0.000421684, 1.61355e-05]
  gravity bias YZ [-0.00111529, 0.000311467]
E5 [33.18, 34.92) s:
  mass x [0.958945, 1.1996, 0.885858, 1.13036]
  CoM XYZ mm [2.48551, -0.882421, 4.8983]
  mu_s [0.704749, 0.538434, 0.546287, 0.568073]
  mu_d [0.541926, 0.538434, 0.546287, 0.568073]
  Kp [steer,rear] [10.4711, 0]; Kd [0.201411, 0]
  spawn roll -0.00859318 rad; rate -0.165972 rad/s
  gyro bias XYZ rad/s [-0.000297346, -0.000132116, 1.4391e-05]
  gravity bias YZ [0.000481221, 0.00282605]
E6 [34.92, 60.00) s:
  mass x [1.15527, 1.01585, 1.04687, 1.03184]
  CoM XYZ mm [-4.60195, 3.53239, -1.15752]
  mu_s [0.700521, 0.568073, 0.616244, 0.528253]
  mu_d [0.641329, 0.568073, 0.586498, 0.528253]
  Kp [steer,rear] [10.6587, 0]; Kd [0.207751, 0]
  spawn roll 0.0648386 rad; rate 0.202985 rad/s
  gyro bias XYZ rad/s [-0.000350997, -3.76622e-05, -2.16671e-05]
  gravity bias YZ [0.00101979, 0.000359558]
Six significant digits shown; full precision: sampled_dr_values.json.
```

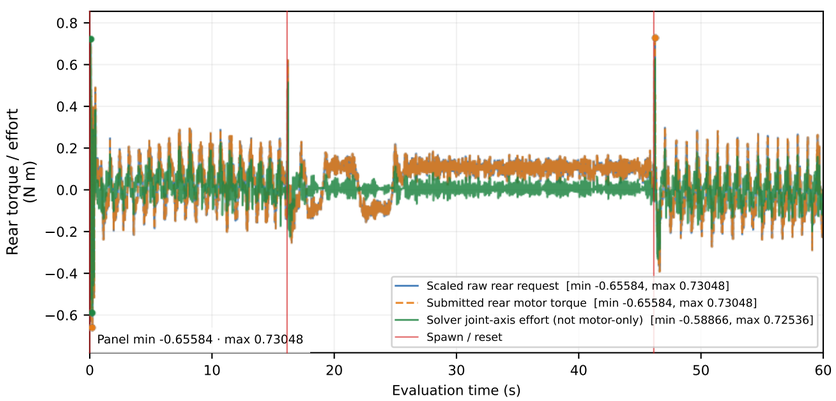
```
SAMPLED DR: env 33; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 1.80) s:
  mass x [0.885317, 0.861252, 1.0657, 1.07373]
  CoM XYZ mm [4.33164, 2.5149, 4.26862]
  mu_s [0.704749, 0.700521, 0.749678, 0.71155]
  mu_d [0.541926, 0.641329, 0.741429, 0.71155]
  Kp [steer,rear] [9.0252, 0]; Kd [0.212922, 0]
  spawn roll -0.0197365 rad; rate 0.0835653 rad/s
  gyro bias XYZ rad/s [-3.26169e-05, 0.000319795, -0.000258576]
  gravity bias YZ [-0.00102961, -0.00178678]
E2 [1.80, 31.80) s:
  mass x [1.07268, 1.1171, 1.18264, 1.04602]
  CoM XYZ mm [-2.91, -4.40165, 2.16322]
  mu_s [0.733515, 0.666375, 0.640867, 0.704749]
  mu_d [0.558392, 0.519145, 0.556841, 0.541926]
  Kp [steer,rear] [9.84797, 0]; Kd [0.1977, 0]
  spawn roll 0.0330717 rad; rate 0.0440185 rad/s
  gyro bias XYZ rad/s [0.00028135, 0.000405887, 4.42249e-05]
  gravity bias YZ [-0.000279101, 0.00258951]
E3 [31.80, 60.00) s:
  mass x [1.04684, 1.16331, 0.83554, 0.81731]
  CoM XYZ mm [-2.19189, 0.128327, 3.89774]
  mu_s [0.611239, 0.611239, 0.640016, 0.732187]
  mu_d [0.611239, 0.611239, 0.513141, 0.631664]
  Kp [steer,rear] [9.59445, 0]; Kd [0.201467, 0]
  spawn roll -0.00351131 rad; rate 0.0643324 rad/s
  gyro bias XYZ rad/s [3.65111e-05, -5.71696e-06, 0.000277037]
  gravity bias YZ [-0.00317783, 0.00182454]
Six significant digits shown; full precision: sampled_dr_values.json.
```

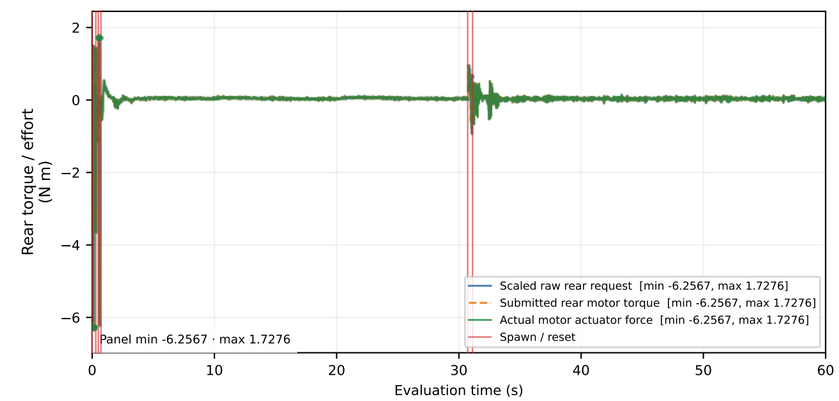
model_2500 · Demand versus actuation · env 34 · full 60 s

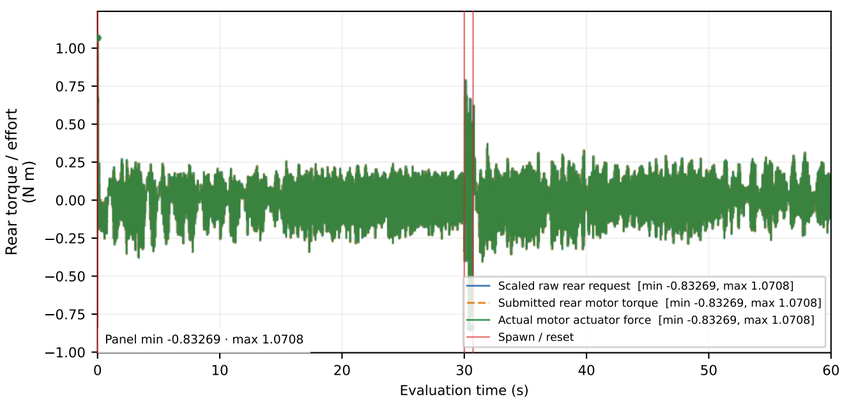












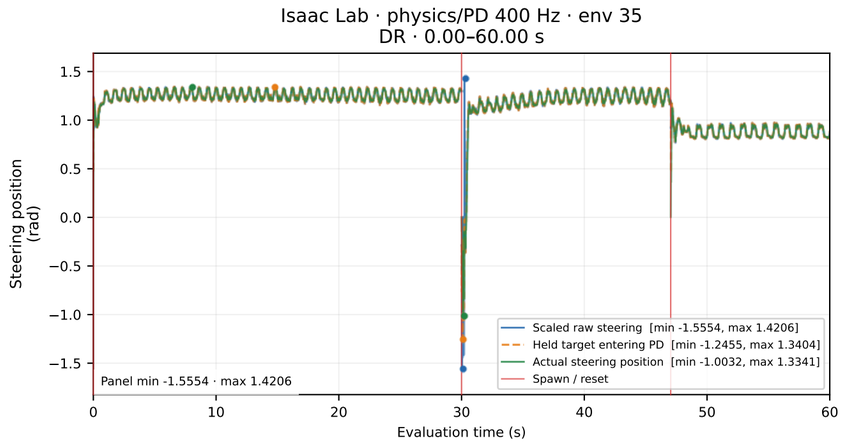
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

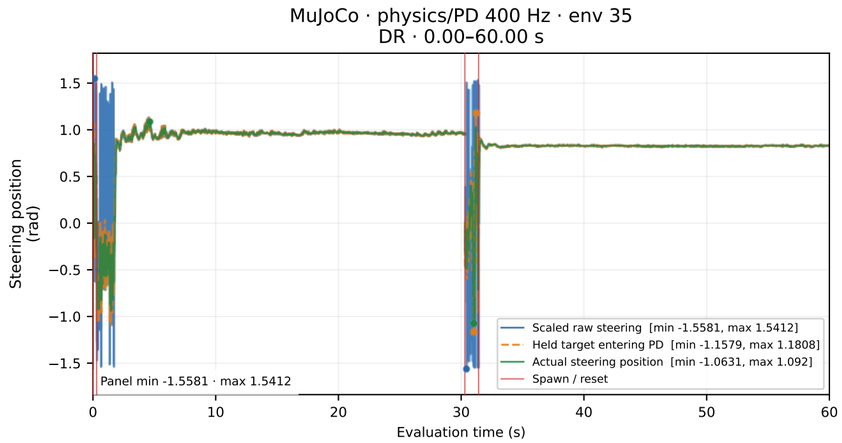
```
SAMPLED DR: env 34; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 16.14) s:
  mass x [0.805827, 1.08268, 0.968777, 1.1818]
  CoM XYZ mm [-4.69837, -2.09229, 0.019148]
  mu_s [0.647391, 0.533455, 0.533455, 0.785466]
  mu_d [0.647391, 0.533455, 0.533455, 0.52258]
  Kp [steer,rear] [9.28387, 0]; Kd [0.215265, 0]
  spawn roll -0.0397326 rad; rate -0.186314 rad/s
  gyro bias XYZ rad/s [-0.000172667, 0.000245817, -0.000265783]
  gravity bias YZ [-0.00288783, -0.000176845]
E2 [16.14, 46.14) s:
  mass x [0.825074, 0.83507, 1.14831, 1.11127]
  CoM XYZ mm [-1.40844, 4.53725, 1.21087]
  mu_s [0.693205, 0.681711, 0.693205, 0.760821]
  mu_d [0.693205, 0.611756, 0.693205, 0.670315]
  Kp [steer,rear] [10.0339, 0]; Kd [0.215451, 0]
  spawn roll 0.0688019 rad; rate -0.0408712 rad/s
  gyro bias XYZ rad/s [-0.000203892, -0.00045729, 0.000102071]
  gravity bias YZ [8.0981e-05, 0.00307239]
E3 [46.14, 60.00) s:
  mass x [1.08489, 1.15281, 1.16129, 1.07297]
  CoM XYZ mm [4.08788, 0.834204, -3.63411]
  mu_s [0.659445, 0.674963, 0.751971, 0.693205]
  mu_d [0.547619, 0.601294, 0.541224, 0.693205]
  Kp [steer,rear] [9.46826, 0]; Kd [0.188964, 0]
  spawn roll 0.0654882 rad; rate -0.035815 rad/s
  gyro bias XYZ rad/s [0.000109645, 8.30663e-05, -0.000334825]
  gravity bias YZ [-0.00258604, -0.000351566]
Six significant digits shown; full precision: sampled_dr_values.json.
```

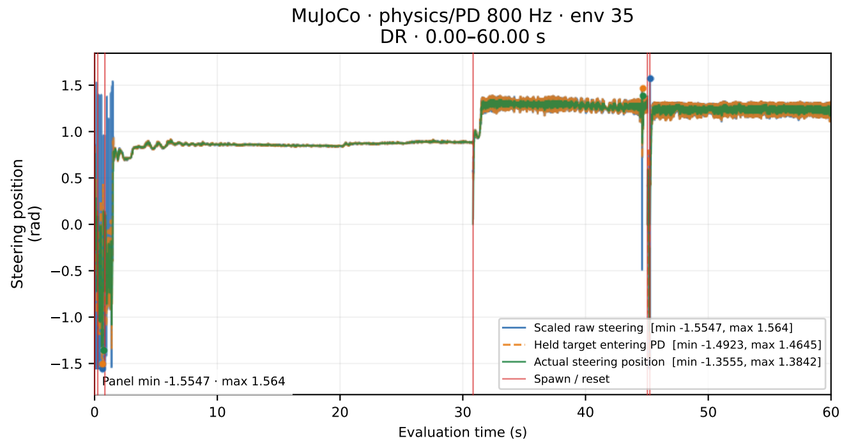
```
SAMPLED DR: env 34; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.30) s:
  mass x [1.12253, 1.19421, 1.07296, 1.08835]
  CoM XYZ mm [-4.21886, -1.91108, 0.617315]
  mu_s [0.666375, 0.700521, 0.637675, 0.633024]
  mu_d [0.519145, 0.641329, 0.637675, 0.568172]
  Kp [steer,rear] [9.69958, 0]; Kd [0.182268, 0]
  spawn roll 0.0739992 rad; rate 0.209117 rad/s
  gyro bias XYZ rad/s [-0.000150689, 0.00024818, -7.63476e-06]
  gravity bias YZ [-0.000274781, -0.000871272]
E2 [0.30, 0.50) s:
  mass x [0.980226, 0.867885, 1.18425, 1.17197]
  CoM XYZ mm [-4.6277, 2.63561, -1.75512]
  mu_s [0.568073, 0.541939, 0.538434, 0.611239]
  mu_d [0.568073, 0.534359, 0.538434, 0.611239]
  Kp [steer,rear] [10.1143, 0]; Kd [0.20075, 0]
  spawn roll 0.0905334 rad; rate 0.248604 rad/s
  gyro bias XYZ rad/s [8.45795e-05, 0.000149024, -4.7732e-05]
  gravity bias YZ [0.00152551, -0.00321815]
E3 [0.50, 0.72) s:
  mass x [0.990933, 0.881057, 1.10269, 1.13758]
  CoM XYZ mm [-0.711965, 1.16799, -3.54662]
  mu_s [0.640867, 0.538434, 0.611239, 0.597748]
  mu_d [0.556841, 0.538434, 0.611239, 0.597748]
  Kp [steer,rear] [10.3673, 0]; Kd [0.201971, 0]
  spawn roll 0.0713888 rad; rate -0.092217 rad/s
  gyro bias XYZ rad/s [-0.000371305, 1.68319e-05, 0.000262002]
  gravity bias YZ [0.00232817, 0.00141465]
E4 [0.72, 30.72) s:
  mass x [1.07908, 1.014, 0.901382, 1.13736]
  CoM XYZ mm [-0.410973, -4.45773, 4.44689]
  mu_s [0.633024, 0.71008, 0.791209, 0.528253]
  mu_d [0.568172, 0.59371, 0.767936, 0.528253]
  Kp [steer,rear] [10.0404, 0]; Kd [0.21151, 0]
  spawn roll -0.0604781 rad; rate 0.24403 rad/s
  gyro bias XYZ rad/s [-8.95758e-07, 0.000374733, -4.32786e-05]
  gravity bias YZ [0.00513053, -0.00115521]
E5 [30.72, 31.12) s:
  mass x [0.994516, 1.11325, 0.884855, 0.872714]
  CoM XYZ mm [-2.46737, 1.57546, 0.486792]
  mu_s [0.568073, 0.700521, 0.666375, 0.631146]
  mu_d [0.568073, 0.641329, 0.519145, 0.631146]
  Kp [steer,rear] [9.88636, 0]; Kd [0.209558, 0]
  spawn roll 0.0833926 rad; rate -0.00718131 rad/s
  gyro bias XYZ rad/s [-0.000106101, -0.000147011, 0.000211347]
  gravity bias YZ [-0.00190917, -0.0003670 94]
E6 [31.12, 60.00) s:
  mass x [0.804439, 1.19971, 1.02108, 0.989968]
  CoM XYZ mm [2.5749, -4.23626, 1.23556]
  mu_s [0.733515, 0.640867, 0.669571, 0.541939]
  mu_d [0.558392, 0.556841, 0.669571, 0.534359]
  Kp [steer,rear] [10.9656, 0]; Kd [0.199435, 0]
  spawn roll -0.0371786 rad; rate -0.178639 rad/s
  gyro bias XYZ rad/s [2.13248e-05, -0.00025547, 0.000179823]
  gravity bias YZ [0.00265764, -0.00570382]
Six significant digits shown; full precision: sampled_dr_values.json.
```

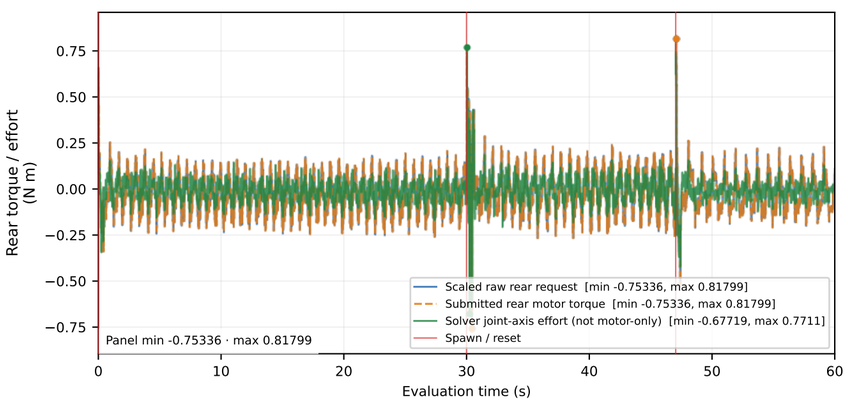
```
SAMPLED DR: env 34; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.12253, 1.19421, 1.07296, 1.08835]
  CoM XYZ mm [-4.21886, -1.91108, 0.617315]
  mu_s [0.666375, 0.700521, 0.637675, 0.633024]
  mu_d [0.519145, 0.641329, 0.637675, 0.568172]
  Kp [steer,rear] [9.69958, 0]; Kd [0.182268, 0]
  spawn roll 0.0739992 rad; rate 0.209117 rad/s
  gyro bias XYZ rad/s [-0.000150689, 0.00024818, -7.63476e-06]
  gravity bias YZ [-0.000274781, -0.000871272]
E2 [30.00, 30.72) s:
  mass x [1.12651, 1.17793, 0.985291, 0.969248]
  CoM XYZ mm [1.6075, 4.6953, -1.06634]
  mu_s [0.640016, 0.732187, 0.728342, 0.71155]
  mu_d [0.513141, 0.631664, 0.728342, 0.71155]
  Kp [steer,rear] [9.58907, 0]; Kd [0.209406, 0]
  spawn roll -0.0759934 rad; rate 0.197094 rad/s
  gyro bias XYZ rad/s [-0.000316628, -0.000359827, -0.000149901]
  gravity bias YZ [-0.00311456, 0.000597333]
E3 [30.72, 60.00) s:
  mass x [0.816499, 0.906458, 0.925283, 0.977816]
  CoM XYZ mm [-2.75219, -1.88555, 3.93467]
  mu_s [0.700521, 0.69316, 0.723429, 0.748289]
  mu_d [0.641329, 0.69316, 0.723429, 0.689499]
  Kp [steer,rear] [9.48051, 0]; Kd [0.20162, 0]
  spawn roll 0.0497188 rad; rate -0.136477 rad/s
  gyro bias XYZ rad/s [-0.000378376, 0.000175181, -7.20776e-05]
  gravity bias YZ [0.00183164, -0.0036954]
Six significant digits shown; full precision: sampled_dr_values.json.
```

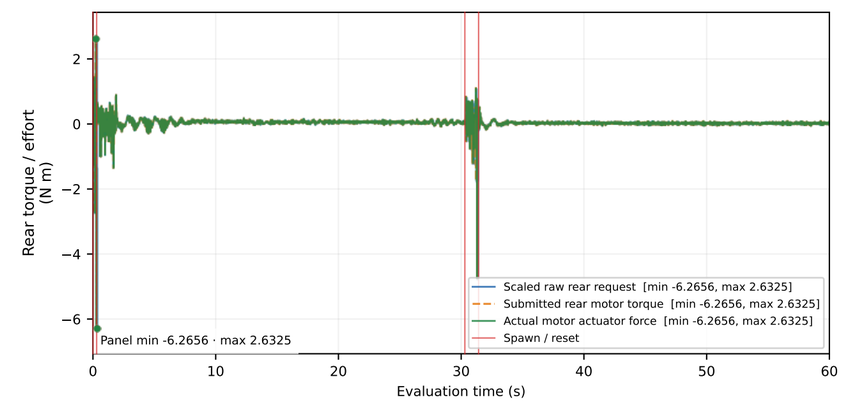
model_2500 · Demand versus actuation · env 35 · full 60 s

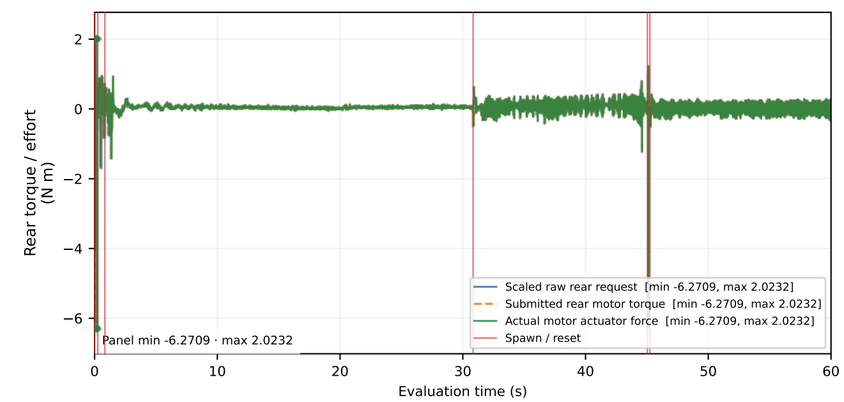












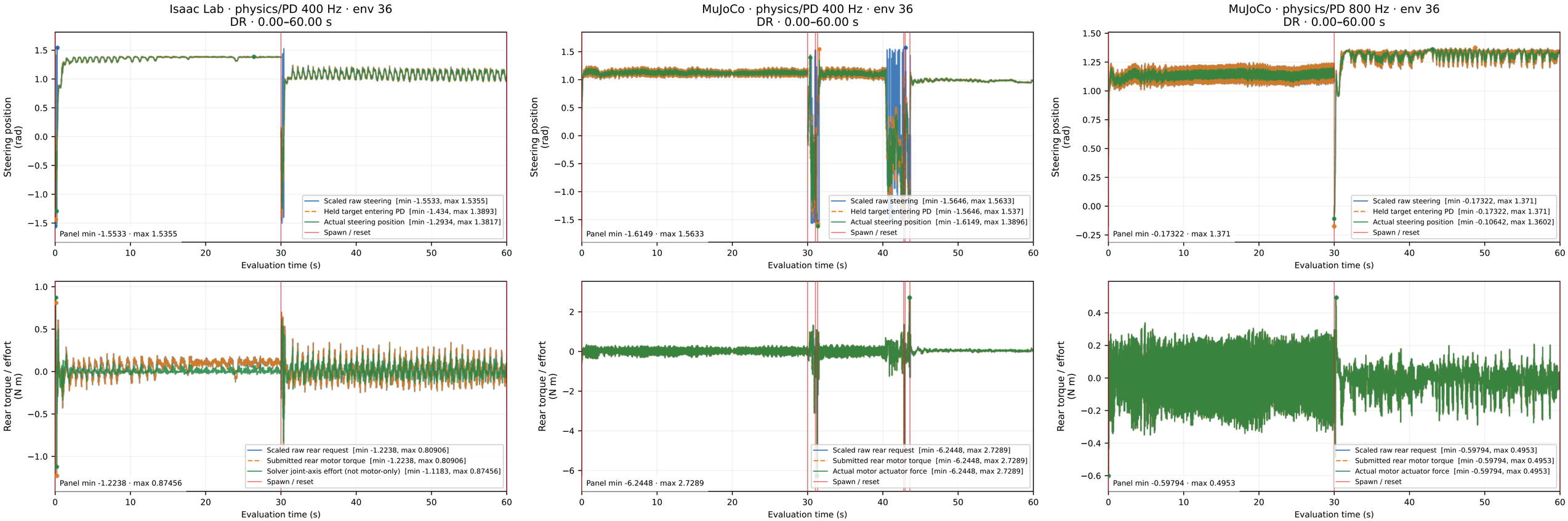
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 35; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.03099, 1.07698, 0.987322, 0.810236]
  CoM XYZ mm [1.15526, -1.85922, -3.17019]
  mu_s [0.751971, 0.532645, 0.774409, 0.547743]
  mu_d [0.541224, 0.532645, 0.561095, 0.547743]
  Kp [steer,rear] [10.4329, 0]; Kd [0.19945, 0]
  spawn roll 0.0616521 rad; rate 0.246993 rad/s
  gyro bias XYZ rad/s [0.000134506, -0.000247672, 0.000167419]
  gravity bias YZ [0.000144644, -0.00120618]
E2 [30.00, 47.04) s:
  mass x [1.16434, 1.18864, 0.980074, 1.1655]
  CoM XYZ mm [-0.537932, -1.72436, -0.138611]
  mu_s [0.659445, 0.547743, 0.785466, 0.774409]
  mu_d [0.547619, 0.547743, 0.52258, 0.561095]
  Kp [steer,rear] [9.36428, 0]; Kd [0.204561, 0]
  spawn roll -0.049918 rad; rate -0.074224 rad/s
  gyro bias XYZ rad/s [4.32197e-05, -0.000233409, 0.000143327]
  gravity bias YZ [-0.00169889, -0.00270366]
E3 [47.04, 60.00) s:
  mass x [0.946494, 0.976625, 1.12899, 0.829175]
  CoM XYZ mm [0.558808, 2.90988, 4.159]
  mu_s [0.576972, 0.532645, 0.751971, 0.533455]
  mu_d [0.576972, 0.532645, 0.541224, 0.533455]
  Kp [steer,rear] [10.5204, 0]; Kd [0.200146, 0]
  spawn roll 0.0970838 rad; rate 0.214616 rad/s
  gyro bias XYZ rad/s [0.000114234, -0.000117443, -0.000127933]
  gravity bias YZ [-0.000617411, -0.000349237]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 35; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.30) s:
  mass x [1.11437, 1.02047, 0.897555, 0.933781]
  CoM XYZ mm [-1.8131, -1.09846, 3.01301]
  mu_s [0.611239, 0.528253, 0.541939, 0.791209]
  mu_d [0.611239, 0.528253, 0.534359, 0.767936]
  Kp [steer,rear] [10.5826, 0]; Kd [0.204159, 0]
  spawn roll 0.0757045 rad; rate -0.169052 rad/s
  gyro bias XYZ rad/s [0.000179138, -0.000372661, -0.000247778]
  gravity bias YZ [0.00193906, -0.00125636]
E2 [0.30, 30.30) s:
  mass x [1.10358, 1.11516, 1.04214, 1.03538]
  CoM XYZ mm [-0.226786, 1.64802, 3.13249]
  mu_s [0.541939, 0.757579, 0.633024, 0.666375]
  mu_d [0.534359, 0.70921, 0.568172, 0.519145]
  Kp [steer,rear] [9.36067, 0]; Kd [0.192032, 0]
  spawn roll -0.0571343 rad; rate 0.00151967 rad/s
  gyro bias XYZ rad/s [0.000262581, -0.000138235, -8.50386e-05]
  gravity bias YZ [-0.00149944, -0.000276668]
E3 [30.30, 31.42) s:
  mass x [0.801902, 1.05794, 1.15258, 0.980549]
  CoM XYZ mm [2.70311, -3.33393, 2.11846]
  mu_s [0.568073, 0.611239, 0.559972, 0.748289]
  mu_d [0.568073, 0.611239, 0.502209, 0.689499]
  Kp [steer,rear] [10.8071, 0]; Kd [0.218447, 0]
  spawn roll -0.0246479 rad; rate -0.2484 rad/s
  gyro bias XYZ rad/s [0.000211986, 0.000154841, 0.000193949]
  gravity bias YZ [-0.00226276, 0.00128962]
E4 [31.42, 60.00) s:
  mass x [1.03001, 0.832794, 0.854795, 1.08168]
  CoM XYZ mm [-1.60549, 4.05211, 3.62333]
  mu_s [0.727426, 0.538434, 0.538976, 0.748289]
  mu_d [0.606358, 0.538434, 0.538976, 0.689499]
  Kp [steer,rear] [9.36101, 0]; Kd [0.214957, 0]
  spawn roll 0.0552535 rad; rate -0.00677227 rad/s
  gyro bias XYZ rad/s [0.000151423, 0.000256563, 0.000106564]
  gravity bias YZ [-0.000977961, 0.00130912]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 35; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.26) s:
  mass x [1.11437, 1.02047, 0.897555, 0.933781]
  CoM XYZ mm [-1.8131, -1.09846, 3.01301]
  mu_s [0.611239, 0.528253, 0.541939, 0.791209]
  mu_d [0.611239, 0.528253, 0.534359, 0.767936]
  Kp [steer,rear] [10.5826, 0]; Kd [0.204159, 0]
  spawn roll 0.0757045 rad; rate -0.169052 rad/s
  gyro bias XYZ rad/s [0.000179138, -0.000372661, -0.000247778]
  gravity bias YZ [0.00193906, -0.00125636]
E2 [0.26, 0.84) s:
  mass x [0.933735, 0.891164, 1.15416, 0.854002]
  CoM XYZ mm [3.52365, 4.83263, -0.209645]
  mu_s [0.71008, 0.633024, 0.640867, 0.538976]
  mu_d [0.59371, 0.568172, 0.556841, 0.538976]
  Kp [steer,rear] [10.2155, 0]; Kd [0.209338, 0]
  spawn roll -0.0365737 rad; rate 0.0966329 rad/s
  gyro bias XYZ rad/s [0.000403267, 8.54849e-05, -0.000120849]
  gravity bias YZ [0.000971239, 0.00115446]
E3 [0.84, 30.84) s:
  mass x [0.889133, 1.09597, 1.1635, 0.92522]
  CoM XYZ mm [0.101439, 3.38965, 4.61652]
  mu_s [0.757579, 0.640867, 0.616244, 0.700521]
  mu_d [0.70921, 0.556841, 0.586498, 0.641329]
  Kp [steer,rear] [10.2071, 0]; Kd [0.206588, 0]
  spawn roll -0.00866833 rad; rate -0.00740132 rad/s
  gyro bias XYZ rad/s [-0.000145499, 4.27028e-05, 0.000216237]
  gravity bias YZ [-0.000840456, -0.00225714]
E4 [30.84, 45.04) s:
  mass x [0.84311, 0.91119, 1.19725, 1.0504]
  CoM XYZ mm [4.53273, -3.95345, -2.3898]
  mu_s [0.616244, 0.757579, 0.732187, 0.538976]
  mu_d [0.586498, 0.70921, 0.631664, 0.538976]
  Kp [steer,rear] [10.1877, 0]; Kd [0.203099, 0]
  spawn roll 0.00957629 rad; rate 0.137029 rad/s
  gyro bias XYZ rad/s [-0.000176587, 0.000154702, -4.8365e-05]
  gravity bias YZ [0.000420363, -0.000894129]
E5 [45.04, 45.24) s:
  mass x [0.871442, 0.948927, 0.998415, 0.833973]
  CoM XYZ mm [-3.79589, -3.47431, -1.76624]
  mu_s [0.631146, 0.640016, 0.700521, 0.733515]
  mu_d [0.631146, 0.513141, 0.641329, 0.558392]
  Kp [steer,rear] [10.3672, 0]; Kd [0.218668, 0]
  spawn roll 0.0871196 rad; rate -0.196338 rad/s
  gyro bias XYZ rad/s [0.000544532, -5.09331e-05, 1.15576e-05]
  gravity bias YZ [0.00155484, -0.00326486]
E6 [45.24, 60.00) s:
  mass x [1.07334, 0.874021, 0.822333, 1.07059]
  CoM XYZ mm [2.44332, -1.95996, 1.66241]
  mu_s [0.538976, 0.559972, 0.728342, 0.568073]
  mu_d [0.538976, 0.502209, 0.728342, 0.568073]
  Kp [steer,rear] [10.1502, 0]; Kd [0.190974, 0]
  spawn roll -0.0383338 rad; rate 0.194893 rad/s
  gyro bias XYZ rad/s [9.6327e-05, -0.00023254, 0.000229655]
  gravity bias YZ [-0.00175814, 0.00257724]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · Demand versus actuation · env 36 · full 60 s



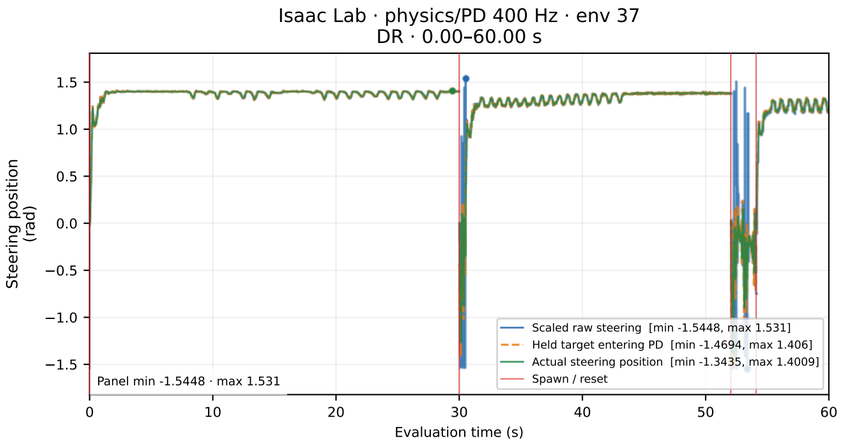
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

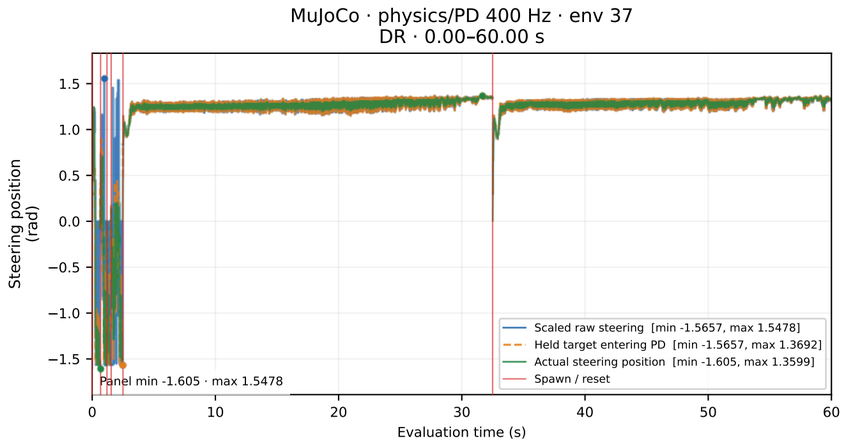
```
SAMPLED DR: env 36; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.889325, 0.823307, 1.18237, 1.18823]
  CoM XYZ mm [-0.652254, -3.03325, -0.461861]
  mu_s [0.794338, 0.579974, 0.533455, 0.764681]
  mu_d [0.526209, 0.579974, 0.533455, 0.764681]
  Kp [steer,rear] [10.0063, 0]; Kd [0.202994, 0]
  spawn roll -0.0838998 rad; rate -0.177664 rad/s
  gyro bias XYZ rad/s [0.000165333, -0.000268149, 0.000176489]
  gravity bias YZ [0.00087258, -0.000498517]
E2 [30.00, 60.00) s:
  mass x [1.11923, 0.87332, 1.10839, 1.15865]
  CoM XYZ mm [-1.55851, 0.599463, -3.3043]
  mu_s [0.787792, 0.632409, 0.703711, 0.598343]
  mu_d [0.617134, 0.589076, 0.703711, 0.598343]
  Kp [steer,rear] [9.84677, 0]; Kd [0.199817, 0]
  spawn roll -0.0626439 rad; rate 0.171068 rad/s
  gyro bias XYZ rad/s [3.1832e-05, 0.000210742, 0.000367422]
  gravity bias YZ [0.000256666, -0.00324526]
Six significant digits shown; full precision: sampled_dr_values.json.
```

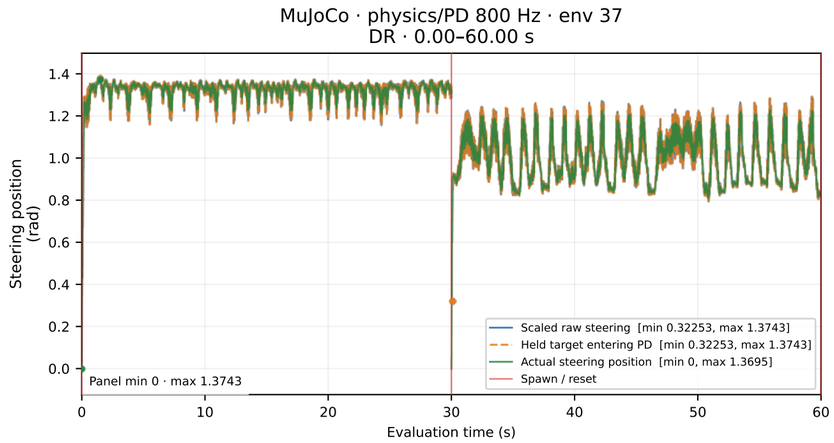
```
SAMPLED DR: env 36; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.967545, 1.16605, 1.00725, 1.13485]
  CoM XYZ mm [-1.27665, 0.695931, -1.20264]
  mu_s [0.538434, 0.727426, 0.631146, 0.538976]
  mu_d [0.538434, 0.606358, 0.631146, 0.538976]
  Kp [steer,rear] [10.3464, 0]; Kd [0.18993, 0]
  spawn roll -0.0134806 rad; rate 0.217073 rad/s
  gyro bias XYZ rad/s [0.000228825, 0.000338871, 0.000202232]
  gravity bias YZ [0.00100058, 0.00282381]
E2 [30.00, 31.04) s:
  mass x [1.19686, 0.970435, 1.07169, 0.896225]
  CoM XYZ mm [-2.67648, -4.26203, 4.45151]
  mu_s [0.728342, 0.611239, 0.748289, 0.669571]
  mu_d [0.728342, 0.611239, 0.689499, 0.669571]
  Kp [steer,rear] [9.8789, 0]; Kd [0.202617, 0]
  spawn roll -0.0216403 rad; rate -0.0927249 rad/s
  gyro bias XYZ rad/s [0.000198632, 0.000211382, -0.000252489]
  gravity bias YZ [-0.000844808, -0.000889235]
E3 [31.04, 31.34) s:
  mass x [1.04945, 0.939586, 1.1852, 0.835938]
  CoM XYZ mm [-0.0678102, 3.15345, 1.75922]
  mu_s [0.559972, 0.559972, 0.700521, 0.616244]
  mu_d [0.502209, 0.502209, 0.641329, 0.586498]
  Kp [steer,rear] [10.8011, 0]; Kd [0.211997, 0]
  spawn roll -0.0658707 rad; rate -0.0579494 rad/s
  gyro bias XYZ rad/s [-2.70543e-05, -0.000117642, -0.00013395]
  gravity bias YZ [-0.00245552, 0.00182166]
E4 [31.34, 42.78) s:
  mass x [0.996467, 0.876334, 0.981981, 1.04004]
  CoM XYZ mm [3.95249, 1.25703, 0.560635]
  mu_s [0.666375, 0.538976, 0.640016, 0.637675]
  mu_d [0.519145, 0.538976, 0.513141, 0.637675]
  Kp [steer,rear] [9.26179, 0]; Kd [0.181723, 0]
  spawn roll 0.0377449 rad; rate 0.0518503 rad/s
  gyro bias XYZ rad/s [2.28952e-05, 8.12965e-05, -6.94963e-05]
  gravity bias YZ [0.00093996, 0.000508482]
E5 [42.78, 42.96) s:
  mass x [0.935606, 0.852864, 1.14186, 1.16011]
  CoM XYZ mm [4.326, 0.230119, 4.46828]
  mu_s [0.700521, 0.541939, 0.637675, 0.791209]
  mu_d [0.641329, 0.534359, 0.637675, 0.767936]
  Kp [steer,rear] [9.06806, 0]; Kd [0.202557, 0]
  spawn roll 0.0745548 rad; rate 0.205111 rad/s
  gyro bias XYZ rad/s [0.000210011, 2.03846e-05, -0.000166352]
  gravity bias YZ [-0.0014714, 0.000873408]
E6 [42.96, 43.60) s:
  mass x [0.958102, 0.841826, 1.16035, 1.14905]
  CoM XYZ mm [4.85714, 2.82715, -4.6075]
  mu_s [0.559972, 0.546287, 0.631146, 0.633024]
  mu_d [0.502209, 0.546287, 0.631146, 0.568172]
  Kp [steer,rear] [10.0447, 0]; Kd [0.197069, 0]
  spawn roll 0.00466166 rad; rate 0.0758323 rad/s
  gyro bias XYZ rad/s [0.000512757, 0.000321531, -0.000107633]
  gravity bias YZ [-0.00185155, -0.000377457]
E7 [43.60, 60.00) s:
  mass x [1.10547, 1.18265, 0.971293, 1.12451]
  CoM XYZ mm [-4.75486, 1.14637, 1.48433]
  mu_s [0.559972, 0.749678, 0.727426, 0.704749]
  mu_d [0.502209, 0.741429, 0.606358, 0.541926]
  Kp [steer,rear] [10.0138, 0]; Kd [0.183276, 0]
  spawn roll 0.0676573 rad; rate -0.114212 rad/s
  gyro bias XYZ rad/s [-0.000322155, -0.000103443, -6.22804e-05]
  gravity bias YZ [0.000823562, 0.001472]
Six significant digits shown; full precision: sampled_dr_values.json.
```

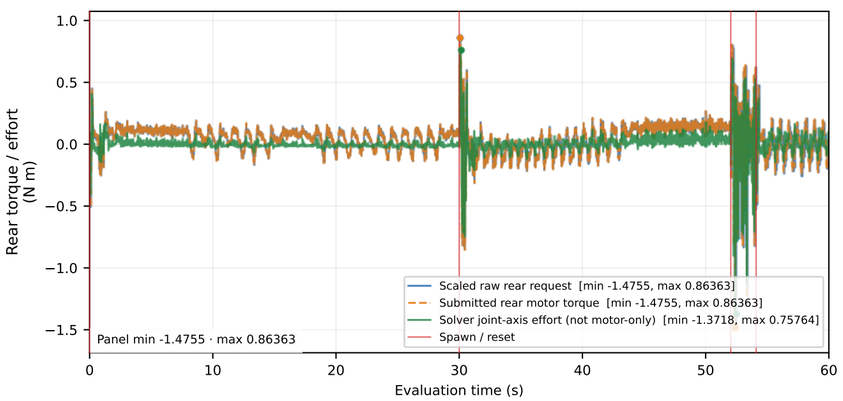
```
SAMPLED DR: env 36; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.967545, 1.16605, 1.00725, 1.13485]
  CoM XYZ mm [-1.27665, 0.695931, -1.20264]
  mu_s [0.538434, 0.727426, 0.631146, 0.538976]
  mu_d [0.538434, 0.606358, 0.631146, 0.538976]
  Kp [steer,rear] [10.3464, 0]; Kd [0.18993, 0]
  spawn roll -0.0134806 rad; rate 0.217073 rad/s
  gyro bias XYZ rad/s [0.000228825, 0.000338871, 0.000202232]
  gravity bias YZ [0.00100058, 0.00282381]
E2 [30.00, 60.00) s:
  mass x [1.19686, 0.970435, 1.07169, 0.896225]
  CoM XYZ mm [-2.67648, -4.26203, 4.45151]
  mu_s [0.728342, 0.611239, 0.748289, 0.669571]
  mu_d [0.728342, 0.611239, 0.689499, 0.669571]
  Kp [steer,rear] [9.8789, 0]; Kd [0.202617, 0]
  spawn roll -0.0216403 rad; rate -0.0927249 rad/s
  gyro bias XYZ rad/s [0.000198632, 0.000211382, -0.000252489]
  gravity bias YZ [-0.000844808, -0.000889235]
Six significant digits shown; full precision: sampled_dr_values.json.
```

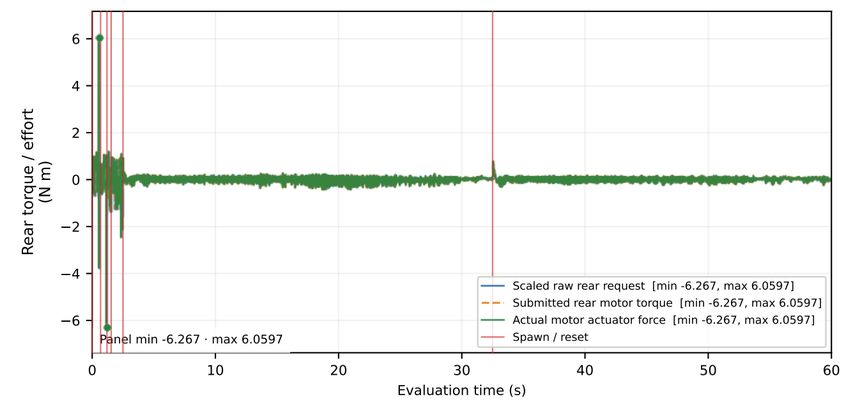
model_2500 · Demand versus actuation · env 37 · full 60 s

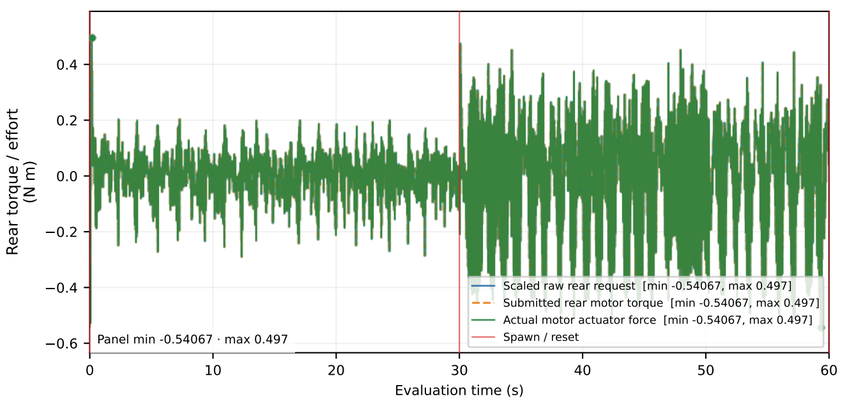












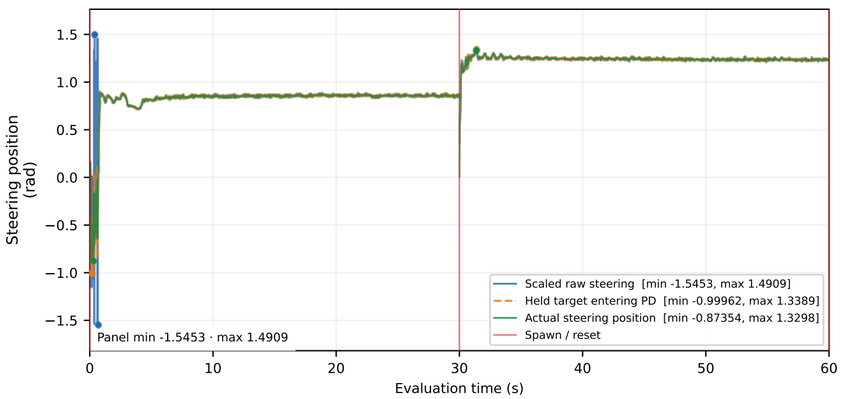
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

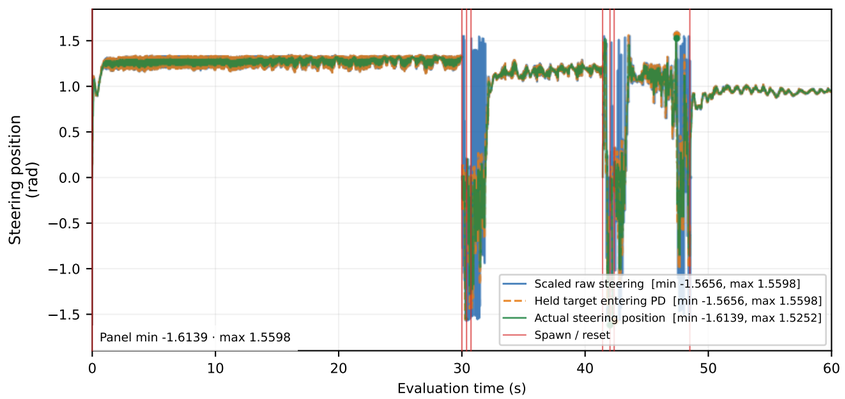
```
SAMPLED DR: env 37; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.06897, 0.86944, 1.04357, 1.07393]
  CoM XYZ mm [1.41931, -3.74473, -4.63834]
  mu_s [0.764681, 0.794338, 0.703711, 0.533455]
  mu_d [0.764681, 0.526209, 0.703711, 0.533455]
  Kp [steer,rear] [10.923, 0]; Kd [0.189328, 0]
  spawn roll -0.0258817 rad; rate -0.00667393 rad/s
  gyro bias XYZ rad/s [0.000218942, -0.00012712, -0.0001508]
  gravity bias YZ [0.00144611, -0.000632127]
E2 [30.00, 52.04) s:
  mass x [0.91121, 0.935715, 1.14339, 1.16952]
  CoM XYZ mm [-3.97393, -2.78473, -1.15326]
  mu_s [0.632409, 0.726175, 0.713129, 0.726175]
  mu_d [0.589076, 0.558574, 0.713129, 0.558574]
  Kp [steer,rear] [10.6151, 0]; Kd [0.211117, 0]
  spawn roll -0.0898541 rad; rate -0.0463745 rad/s
  gyro bias XYZ rad/s [-1.3226e-05, -8.08991e-06, -0.000160178]
  gravity bias YZ [0.00282522, -0.000587555]
E3 [52.04, 54.10) s:
  mass x [1.00566, 1.10026, 0.903106, 1.14291]
  CoM XYZ mm [0.992984, 3.46284, 4.27411]
  mu_s [0.673378, 0.533455, 0.659445, 0.533455]
  mu_d [0.673378, 0.533455, 0.547619, 0.533455]
  Kp [steer,rear] [9.40009, 0]; Kd [0.181659, 0]
  spawn roll -0.0688357 rad; rate 0.143508 rad/s
  gyro bias XYZ rad/s [9.38244e-05, -0.00012617, 0.000102573]
  gravity bias YZ [0.0019121, 0.000154381]
E4 [54.10, 60.00) s:
  mass x [1.12016, 0.933536, 0.905927, 1.11461]
  CoM XYZ mm [-0.808045, -1.85707, -0.963807]
  mu_s [0.632409, 0.602694, 0.632409, 0.602694]
  mu_d [0.589076, 0.602694, 0.589076, 0.602694]
  Kp [steer,rear] [9.15919, 0]; Kd [0.20601, 0]
  spawn roll -0.00417519 rad; rate -0.114425 rad/s
  gyro bias XYZ rad/s [0.000289598, -0.000254552, 0.00039431]
  gravity bias YZ [-0.00250523, -0.00284714]
Six significant digits shown; full precision: sampled_dr_values.json.
```

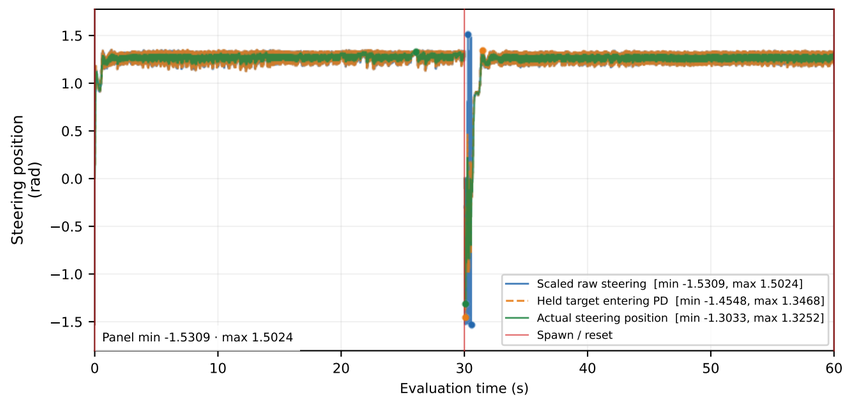
```
SAMPLED DR: env 37; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.68) s:
  mass x [1.16784, 0.828649, 1.16923, 0.814299]
  CoM XYZ mm [-4.12528, -3.74298, -0.222992]
  mu_s [0.749678, 0.791209, 0.616244, 0.728342]
  mu_d [0.741429, 0.767936, 0.586498, 0.728342]
  Kp [steer,rear] [10.2426, 0]; Kd [0.204752, 0]
  spawn roll -0.0450624 rad; rate 0.221879 rad/s
  gyro bias XYZ rad/s [0.000112079, -9.4086e-05, 0.00014188]
  gravity bias YZ [2.36296e-05, 0.00159927]
E2 [0.68, 1.20) s:
  mass x [1.04252, 1.02903, 0.811093, 0.871413]
  CoM XYZ mm [3.17014, -4.5089, 0.716271]
  mu_s [0.736077, 0.640867, 0.637675, 0.733515]
  mu_d [0.699455, 0.556841, 0.637675, 0.558392]
  Kp [steer,rear] [10.7549, 0]; Kd [0.216135, 0]
  spawn roll -0.0756035 rad; rate 0.107357 rad/s
  gyro bias XYZ rad/s [-6.14345e-05, -0.000103861, 0.000441419]
  gravity bias YZ [0.000703019, -0.000120722]
E3 [1.20, 1.54) s:
  mass x [0.842745, 1.10896, 1.10692, 1.01941]
  CoM XYZ mm [-4.18722, 4.37099, 4.20679]
  mu_s [0.597748, 0.640016, 0.568073, 0.727426]
  mu_d [0.597748, 0.513141, 0.568073, 0.606358]
  Kp [steer,rear] [9.37334, 0]; Kd [0.210042, 0]
  spawn roll -0.0998609 rad; rate 0.142483 rad/s
  gyro bias XYZ rad/s [3.11612e-05, -0.00021694, -0.00016323]
  gravity bias YZ [-0.00213008, -0.000319505]
E4 [1.54, 2.50) s:
  mass x [0.877247, 1.03203, 0.908988, 0.877389]
  CoM XYZ mm [-0.511892, -0.395673, 1.73933]
  mu_s [0.700521, 0.700521, 0.541939, 0.616244]
  mu_d [0.641329, 0.641329, 0.534359, 0.586498]
  Kp [steer,rear] [9.50982, 0]; Kd [0.188853, 0]
  spawn roll -0.0598529 rad; rate 0.138223 rad/s
  gyro bias XYZ rad/s [-0.000230918, -0.000146542, 7.09895e-05]
  gravity bias YZ [-0.000158763, -0.00332722]
E5 [2.50, 32.50) s:
  mass x [1.12477, 1.18193, 0.930487, 1.10288]
  CoM XYZ mm [-2.94233, -3.764, -0.874708]
  mu_s [0.727426, 0.748289, 0.611239, 0.704749]
  mu_d [0.606358, 0.689499, 0.611239, 0.541926]
  Kp [steer,rear] [9.46844, 0]; Kd [0.185555, 0]
  spawn roll 0.0677484 rad; rate 0.188911 rad/s
  gyro bias XYZ rad/s [5.97027e-05, -7.08922e-05, 8.33852e-05]
  gravity bias YZ [0.00199517, 0.00348367]
E6 [32.50, 60.00) s:
  mass x [1.17385, 1.18431, 1.02134, 1.13001]
  CoM XYZ mm [-2.76365, -3.50858, 0.862396]
  mu_s [0.611239, 0.700521, 0.728342, 0.732187]
  mu_d [0.611239, 0.641329, 0.728342, 0.631664]
  Kp [steer,rear] [10.734, 0]; Kd [0.19728, 0]
  spawn roll 0.0552036 rad; rate -0.142578 rad/s
  gyro bias XYZ rad/s [-0.000333262, -0.000305812, -0.000206869]
  gravity bias YZ [0.000398298, -0.00223014]
Six significant digits shown; full precision: sampled_dr_values.json.
```

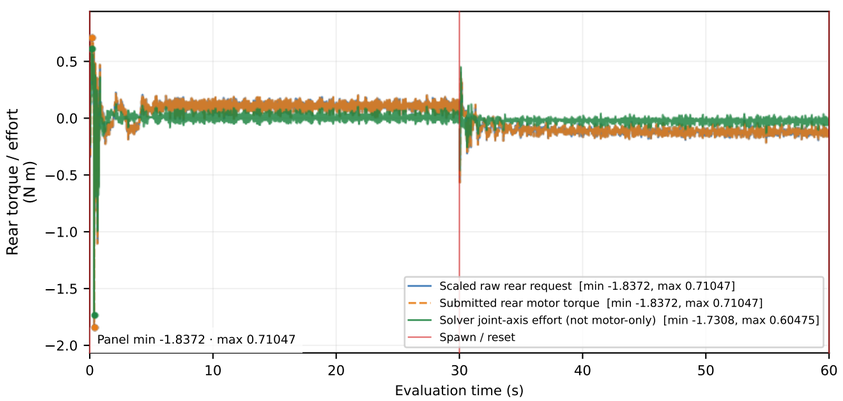
```
SAMPLED DR: env 37; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.16784, 0.828649, 1.16923, 0.814299]
  CoM XYZ mm [-4.12528, -3.74298, -0.222992]
  mu_s [0.749678, 0.791209, 0.616244, 0.728342]
  mu_d [0.741429, 0.767936, 0.586498, 0.728342]
  Kp [steer,rear] [10.2426, 0]; Kd [0.204752, 0]
  spawn roll -0.0450624 rad; rate 0.221879 rad/s
  gyro bias XYZ rad/s [0.000112079, -9.4086e-05, 0.00014188]
  gravity bias YZ [2.36296e-05, 0.00159927]
E2 [30.00, 60.00) s:
  mass x [0.826023, 1.04052, 0.930369, 0.969218]
  CoM XYZ mm [1.75077, 2.31945, 3.03961]
  mu_s [0.541939, 0.71008, 0.637675, 0.597748]
  mu_d [0.534359, 0.59371, 0.637675, 0.597748]
  Kp [steer,rear] [10.4509, 0]; Kd [0.200802, 0]
  spawn roll 0.0432316 rad; rate -0.0721844 rad/s
  gyro bias XYZ rad/s [2.86129e-05, -0.00023106, 2.02665e-05]
  gravity bias YZ [0.00152038, 0.000273868]
Six significant digits shown; full precision: sampled_dr_values.json.
```

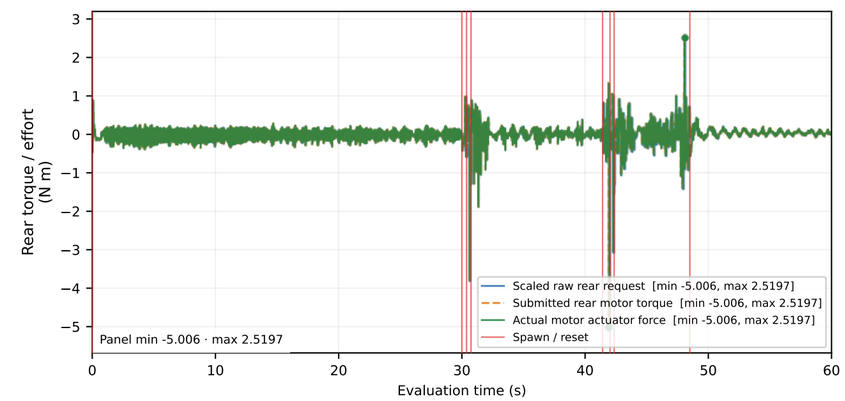
model_2500 · Demand versus actuation · env 38 · full 60 s

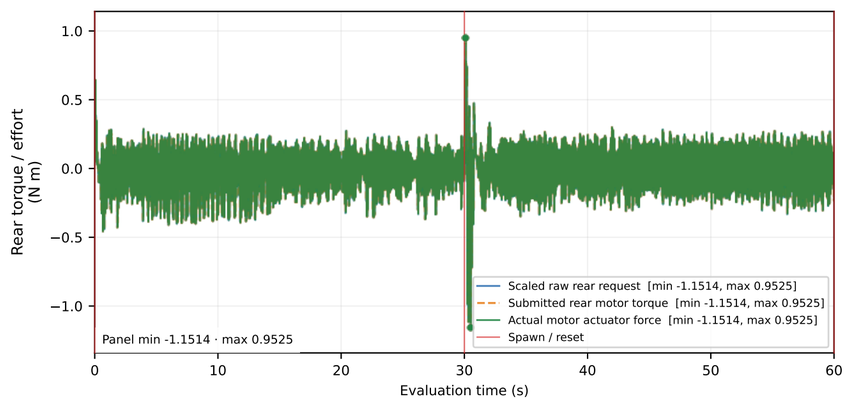












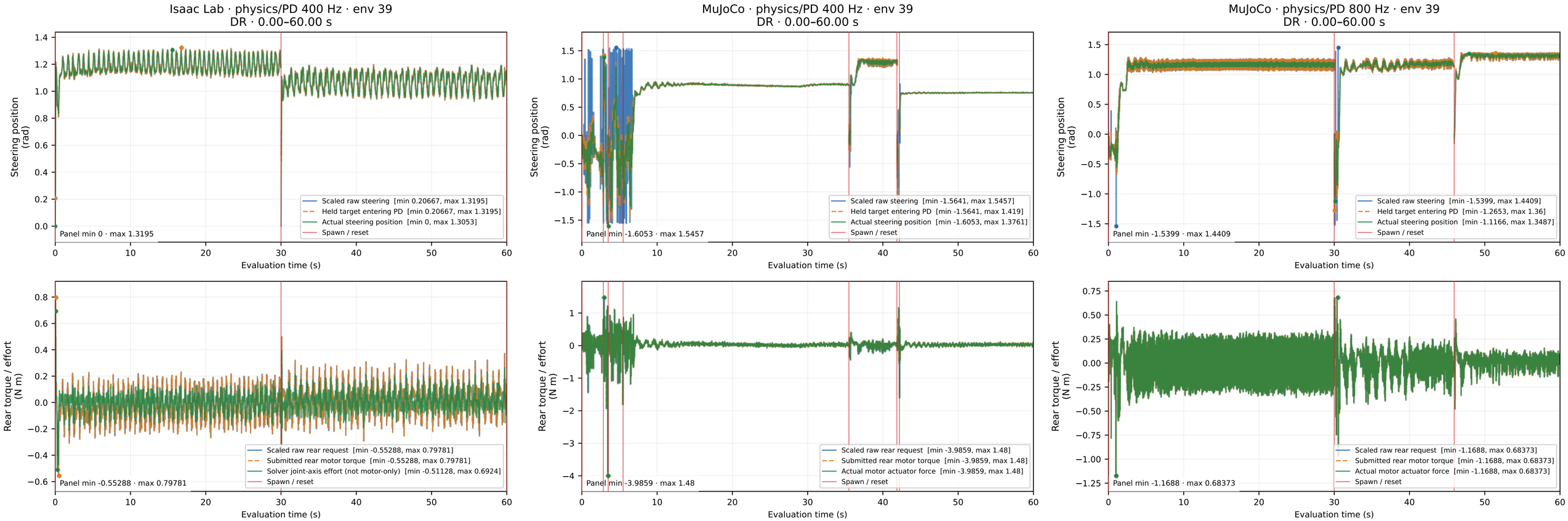
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 38; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.11762, 0.836914, 0.995436, 1.1308]
  CoM XYZ mm [3.84191, 4.0777, -4.95727]
  mu_s [0.794338, 0.647391, 0.673378, 0.531594]
  mu_d [0.526209, 0.647391, 0.673378, 0.531594]
  Kp [steer,rear] [10.4319, 0]; Kd [0.194748, 0]
  spawn roll -0.011885 rad; rate 0.0115525 rad/s
  gyro bias XYZ rad/s [-9.9337e-05, -9.97786e-05, -0.000188746]
  gravity bias YZ [-0.00060973, -0.00307423]
E2 [30.00, 60.00) s:
  mass x [0.948215, 0.951209, 0.903158, 1.15621]
  CoM XYZ mm [1.84169, -2.7413, 3.49542]
  mu_s [0.547743, 0.774409, 0.560535, 0.673378]
  mu_d [0.547743, 0.561095, 0.560535, 0.673378]
  Kp [steer,rear] [9.75346, 0]; Kd [0.181468, 0]
  spawn roll -0.0282039 rad; rate 0.14784 rad/s
  gyro bias XYZ rad/s [-3.54838e-05, -1.54418e-05, -8.73341e-05]
  gravity bias YZ [-0.00133313, 0.0021975]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 38; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.0946, 0.807368, 0.814653, 1.03788]
  CoM XYZ mm [0.857136, -2.76032, 1.38134]
  mu_s [0.736077, 0.704749, 0.633024, 0.616244]
  mu_d [0.699455, 0.541926, 0.568172, 0.586498]
  Kp [steer,rear] [9.30934, 0]; Kd [0.192084, 0]
  spawn roll 0.0494247 rad; rate -0.184587 rad/s
  gyro bias XYZ rad/s [-0.000134545, 0.000185694, -0.000102061]
  gravity bias YZ [-0.00183624, -0.000511672]
E2 [30.00, 30.38) s:
  mass x [0.890986, 1.19204, 0.890429, 0.852831]
  CoM XYZ mm [1.17917, -2.10881, 4.54459]
  mu_s [0.748289, 0.640016, 0.528253, 0.631146]
  mu_d [0.689499, 0.513141, 0.528253, 0.631146]
  Kp [steer,rear] [9.76925, 0]; Kd [0.187092, 0]
  spawn roll -0.0858828 rad; rate 0.0235947 rad/s
  gyro bias XYZ rad/s [-0.000119004, -7.4116e-05, -7.19028e-05]
  gravity bias YZ [0.00206062, 0.000455609]
E3 [30.38, 30.74) s:
  mass x [0.81892, 0.846206, 1.14164, 0.823848]
  CoM XYZ mm [-4.11037, 0.686048, 3.32889]
  mu_s [0.640867, 0.538434, 0.538434, 0.704749]
  mu_d [0.556841, 0.538434, 0.538434, 0.541926]
  Kp [steer,rear] [9.50962, 0]; Kd [0.202056, 0]
  spawn roll -0.0384562 rad; rate 0.0829481 rad/s
  gyro bias XYZ rad/s [-3.30915e-05, -0.00019653, -1.83674e-05]
  gravity bias YZ [0.000780591, 0.00202102]
E4 [30.74, 41.42) s:
  mass x [1.06578, 0.990997, 1.04596, 0.891202]
  CoM XYZ mm [-4.88459, -0.747627, -0.747745]
  mu_s [0.728342, 0.704749, 0.733515, 0.704749]
  mu_d [0.728342, 0.541926, 0.558392, 0.541926]
  Kp [steer,rear] [10.4303, 0]; Kd [0.194417, 0]
  spawn roll -0.0704001 rad; rate -0.197332 rad/s
  gyro bias XYZ rad/s [-2.40726e-05, 2.11562e-05, 0.000105931]
  gravity bias YZ [0.000769275, -0.000822098]
E5 [41.42, 42.02) s:
  mass x [1.12862, 0.860983, 0.910753, 0.814961]
  CoM XYZ mm [-2.81092, -2.96765, -3.40159]
  mu_s [0.748289, 0.546287, 0.538976, 0.723429]
  mu_d [0.689499, 0.546287, 0.538976, 0.723429]
  Kp [steer,rear] [10.7562, 0]; Kd [0.200965, 0]
  spawn roll 0.0476956 rad; rate 0.0818871 rad/s
  gyro bias XYZ rad/s [-0.000138082, 0.000215344, 0.000224316]
  gravity bias YZ [-0.00227307, 0.000391213]
E6 [42.02, 42.36) s:
  mass x [1.17562, 1.12714, 0.899412, 0.975255]
  CoM XYZ mm [-0.540265, -0.367724, -3.62225]
  mu_s [0.791209, 0.71008, 0.669571, 0.528253]
  mu_d [0.767936, 0.59371, 0.669571, 0.528253]
  Kp [steer,rear] [9.98523, 0]; Kd [0.210272, 0]
  spawn roll -0.0555902 rad; rate -0.107512 rad/s
  gyro bias XYZ rad/s [7.89347e-05, 0.000265749, 0.000297908]
  gravity bias YZ [-0.0043962, 0.00136375]
E7 [42.36, 48.50) s:
  mass x [0.863007, 1.19554, 0.975987, 0.931001]
  CoM XYZ mm [0.43362, -0.380969, -4.3105]
  mu_s [0.666375, 0.749678, 0.71008, 0.633024]
  mu_d [0.519145, 0.741429, 0.59371, 0.568172]
  Kp [steer,rear] [9.91488, 0]; Kd [0.197156, 0]
  spawn roll 0.0026603 rad; rate -0.145111 rad/s
  gyro bias XYZ rad/s [0.00036954, 0.000367114, -0.000111937]
  gravity bias YZ [-0.000991216, -0.00120392]
E8 [48.50, 60.00) s:
  mass x [1.00666, 1.18752, 0.84588, 0.946555]
  CoM XYZ mm [-4.85837, 1.58286, 3.49196]
  mu_s [0.538976, 0.748289, 0.728342, 0.546287]
  mu_d [0.538976, 0.689499, 0.728342, 0.546287]
  Kp [steer,rear] [10.0283, 0]; Kd [0.188856, 0]
  spawn roll -0.0358756 rad; rate 0.207634 rad/s
  gyro bias XYZ rad/s [-0.000185577, -9.61102e-05, -7.92296e-06]
  gravity bias YZ [-0.000677098, -0.000470997]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 38; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.0946, 0.807368, 0.814653, 1.03788]
  CoM XYZ mm [0.857136, -2.76032, 1.38134]
  mu_s [0.736077, 0.704749, 0.633024, 0.616244]
  mu_d [0.699455, 0.541926, 0.568172, 0.586498]
  Kp [steer,rear] [9.30934, 0]; Kd [0.192084, 0]
  spawn roll 0.0494247 rad; rate -0.184587 rad/s
  gyro bias XYZ rad/s [-0.000134545, 0.000185694, -0.000102061]
  gravity bias YZ [-0.00183624, -0.000511672]
E2 [30.00, 60.00) s:
  mass x [0.890986, 1.19204, 0.890429, 0.852831]
  CoM XYZ mm [1.17917, -2.10881, 4.54459]
  mu_s [0.748289, 0.640016, 0.528253, 0.631146]
  mu_d [0.689499, 0.513141, 0.528253, 0.631146]
  Kp [steer,rear] [9.76925, 0]; Kd [0.187092, 0]
  spawn roll -0.0858828 rad; rate 0.0235947 rad/s
  gyro bias XYZ rad/s [-0.000119004, -7.4116e-05, -7.19028e-05]
  gravity bias YZ [0.00206062, 0.000455609]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · Demand versus actuation · env 39 · full 60 s



50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 39; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.18107, 0.976622, 1.16125, 0.85314]
  CoM XYZ mm [0.932112, -1.2884, 0.629038]
  mu_s [0.576972, 0.579974, 0.628821, 0.632409]
  mu_d [0.576972, 0.579974, 0.628821, 0.589076]
  Kp [steer,rear] [10.5127, 0]; Kd [0.189964, 0]
  spawn roll 0.0735025 rad; rate -0.229389 rad/s
  gyro bias XYZ rad/s [-7.55464e-05, -0.000265604, 0.000107245]
  gravity bias YZ [-0.00155328, -0.000994103]
E2 [30.00, 60.00) s:
  mass x [0.808221, 0.855749, 1.13878, 1.12369]
  CoM XYZ mm [-3.80968, 0.309244, 1.10134]
  mu_s [0.693205, 0.760821, 0.751971, 0.674963]
  mu_d [0.693205, 0.670315, 0.541224, 0.601294]
  Kp [steer,rear] [10.7521, 0]; Kd [0.189601, 0]
  spawn roll 0.0190763 rad; rate 0.0728261 rad/s
  gyro bias XYZ rad/s [2.52376e-07, 0.000105377, 0.000172258]
  gravity bias YZ [-0.00202497, -0.000776156]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 39; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 2.86) s:
  mass x [0.973645, 0.884651, 1.16481, 1.04419]
  CoM XYZ mm [-0.554547, 0.030713, -0.346279]
  mu_s [0.736077, 0.728342, 0.640867, 0.666375]
  mu_d [0.699455, 0.728342, 0.556841, 0.519145]
  Kp [steer,rear] [9.09012, 0]; Kd [0.208688, 0]
  spawn roll -0.00991366 rad; rate -0.150883 rad/s
  gyro bias XYZ rad/s [-0.00036369, 8.10202e-05, 0.000139191]
  gravity bias YZ [0.000357033, 0.00113532]
E2 [2.86, 3.50) s:
  mass x [1.15148, 0.949854, 1.14238, 0.878367]
  CoM XYZ mm [4.51665, -2.24625, 4.09581]
  mu_s [0.559972, 0.69316, 0.704749, 0.71008]
  mu_d [0.502209, 0.69316, 0.541926, 0.59371]
  Kp [steer,rear] [10.3539, 0]; Kd [0.212476, 0]
  spawn roll 0.0796308 rad; rate 0.135644 rad/s
  gyro bias XYZ rad/s [-5.05106e-06, -0.000476714, -0.000228683]
  gravity bias YZ [0.00160196, 0.00495476]
E3 [3.50, 5.48) s:
  mass x [1.09439, 1.02436, 1.12288, 0.804516]
  CoM XYZ mm [-2.56357, -0.297027, 1.67533]
  mu_s [0.669571, 0.666375, 0.732187, 0.637675]
  mu_d [0.669571, 0.519145, 0.631664, 0.637675]
  Kp [steer,rear] [9.88587, 0]; Kd [0.184779, 0]
  spawn roll -0.0288558 rad; rate -0.0109164 rad/s
  gyro bias XYZ rad/s [8.3864e-05, 0.000309233, 0.000301804]
  gravity bias YZ [-0.00292473, -0.00174857]
E4 [5.48, 35.48) s:
  mass x [1.1337, 1.19379, 0.812131, 0.829387]
  CoM XYZ mm [4.27955, 3.12757, 1.81693]
  mu_s [0.704749, 0.640016, 0.568073, 0.704749]
  mu_d [0.541926, 0.513141, 0.568073, 0.541926]
  Kp [steer,rear] [10.579, 0]; Kd [0.182887, 0]
  spawn roll -0.0310055 rad; rate -0.235178 rad/s
  gyro bias XYZ rad/s [-0.000155919, -0.000381763, -0.000303682]
  gravity bias YZ [-0.000194632, 0.00101622]
E5 [35.48, 41.86) s:
  mass x [0.824867, 1.07223, 1.09368, 1.08353]
  CoM XYZ mm [-1.38669, -4.9189, -0.519183]
  mu_s [0.616244, 0.69316, 0.71155, 0.71155]
  mu_d [0.586498, 0.69316, 0.71155, 0.71155]
  Kp [steer,rear] [10.3698, 0]; Kd [0.182396, 0]
  spawn roll -0.0126218 rad; rate -0.140776 rad/s
  gyro bias XYZ rad/s [0.000145366, -0.000175, 0.000220368]
  gravity bias YZ [0.0015459, -6.49389e-05]
E6 [41.86, 42.20) s:
  mass x [0.857126, 1.01041, 0.928453, 1.01033]
  CoM XYZ mm [4.65481, -4.02962, 2.75011]
  mu_s [0.559972, 0.727426, 0.71008, 0.546287]
  mu_d [0.502209, 0.606358, 0.59371, 0.546287]
  Kp [steer,rear] [10.4123, 0]; Kd [0.199519, 0]
  spawn roll -0.0852302 rad; rate 0.0692869 rad/s
  gyro bias XYZ rad/s [5.6216e-05, -0.000164477, 6.30443e-06]
  gravity bias YZ [-0.00309446, -0.00183361]
E7 [42.20, 60.00) s:
  mass x [1.05785, 0.908579, 0.876384, 0.854862]
  CoM XYZ mm [-2.4568, 4.36928, -0.17605]
  mu_s [0.791209, 0.732187, 0.631146, 0.736077]
  mu_d [0.767936, 0.631664, 0.631146, 0.699455]
  Kp [steer,rear] [10.2709, 0]; Kd [0.180311, 0]
  spawn roll 0.0450392 rad; rate -0.187041 rad/s
  gyro bias XYZ rad/s [0.000214484, 0.000120642, 0.000102376]
  gravity bias YZ [-0.00178587, 0.00240232]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 39; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.973645, 0.884651, 1.16481, 1.04419]
  CoM XYZ mm [-0.554547, 0.030713, -0.346279]
  mu_s [0.736077, 0.728342, 0.640867, 0.666375]
  mu_d [0.699455, 0.728342, 0.556841, 0.519145]
  Kp [steer,rear] [9.09012, 0]; Kd [0.208688, 0]
  spawn roll -0.00991366 rad; rate -0.150883 rad/s
  gyro bias XYZ rad/s [-0.00036369, 8.10202e-05, 0.000139191]
  gravity bias YZ [0.000357033, 0.00113532]
E2 [30.00, 45.94) s:
  mass x [1.15842, 1.15223, 0.848549, 0.849509]
  CoM XYZ mm [-2.43123, -0.257908, 1.59667]
  mu_s [0.748289, 0.704749, 0.727426, 0.546287]
  mu_d [0.689499, 0.541926, 0.606358, 0.546287]
  Kp [steer,rear] [9.20614, 0]; Kd [0.215893, 0]
  spawn roll -0.0622703 rad; rate -0.0466344 rad/s
  gyro bias XYZ rad/s [0.000104251, 0.000172304, 0.000264019]
  gravity bias YZ [0.0030847, -0.00269552]
E3 [45.94, 60.00) s:
  mass x [1.1343, 1.00338, 1.0844, 1.03676]
  CoM XYZ mm [-2.87866, -4.3334, 0.3647467]
  mu_s [0.749678, 0.757579, 0.546287, 0.704749]
  mu_d [0.741429, 0.70921, 0.546287, 0.541926]
  Kp [steer,rear] [9.59958, 0]; Kd [0.206538, 0]
  spawn roll -0.0196737 rad; rate -0.0922485 rad/s
  gyro bias XYZ rad/s [-0.000267279, 0.000200594, 0.00017391]
  gravity bias YZ [-0.00165565, -0.00111979]
Six significant digits shown; full precision: sampled_dr_values.json.
```